



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Wélbster Florentino Labat Uchôas
December 31, 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Project Goal:** Predict the successful landing of the Falcon 9 first stage to estimate launch costs using data science techniques.
- **Methodology:**
 - Data Collection:** Extracted data using SpaceX API and Web Scraping (Wikipedia).
 - EDA & SQL:** Performed data cleaning and exploratory analysis to identify patterns in launch sites, payload mass, and orbit types.
 - Visualization:** Created interactive maps (Folium) and a dynamic dashboard (Plotly Dash) for geospatial and trend analysis.
 - Prediction:** Built and tuned classification models (Logistic Regression, SVM, Decision Tree, KNN).
- ° **Key Results:**
 - Launch success rate has significantly increased over time.
 - Heavier payloads and GTO orbits have lower success rates.
 - KSC LC-39A is the most frequent launch site.
 - The predictive model achieved an accuracy of **0.83 (83.33%)**.

Introduction

- **Project Background:**

- SpaceX has revolutionized the space industry with reusable rockets.
- A standard launch costs 62 million dollars.
- This is significantly cheaper than competitors (often over 165 million dollars).
- The key is the recovery of the Falcon 9 first stage.

- **Problem to Solve:**

- The goal is to predict if the first stage will land successfully.
- This prediction helps estimate the true cost of a launch.
- This insight allows other companies to better compete for contracts against SpaceX.



Section 1

Methodology

Methodology

Executive Summary

- **Data collection methodology:**

- Used SpaceX REST API to get launch data.
- Used Web Scraping from Wikipedia to get Falcon 9 launch records.

- **Perform data wrangling:**

- Filtered data for Falcon 9 launches only.
- Handled missing values (e.g., replaced missing payload mass with the mean).
- Converted categorical variables using One-Hot Encoding.

- **Perform exploratory data analysis (EDA) using visualization and SQL:**

- Used SQL queries to calculate success rates and payloads.
- Created charts with Matplotlib/Seaborn to visualize trends (e.g., success rate by orbit).

Methodology

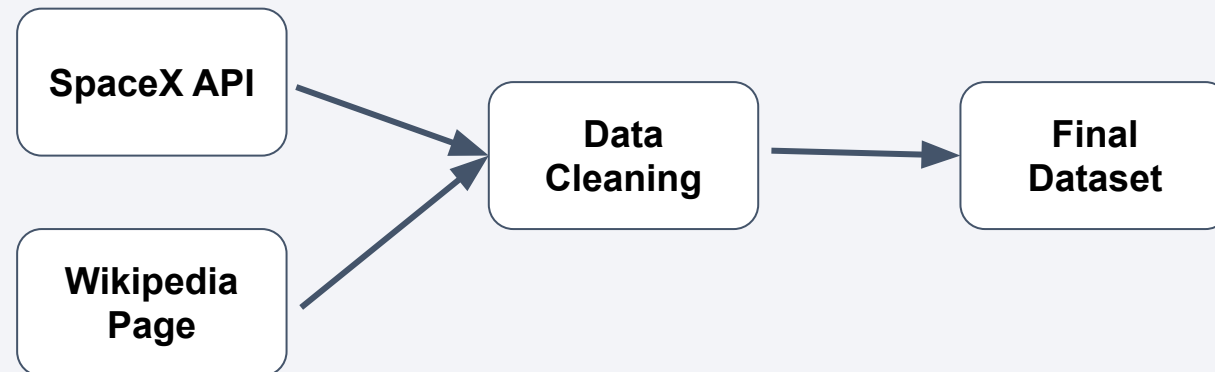
Executive Summary

- **Perform interactive visual analytics using Folium and Plotly Dash:**
 - Built interactive maps with Folium to visualize launch sites and landing outcomes.
 - Created a Plotly Dash dashboard for dynamic data analysis (Payload vs. Success).
- **Perform predictive analysis using classification models:**
 - Split data into training and testing sets.
 - Built and tuned models: Logistic Regression, SVM, Decision Tree, and KNN.
 - Used GridSearchCV for hyperparameter tuning to find the best model.

Data Collection

- **Data Collection Sources:**

- SpaceX API: Used requests library to fetch launch data (rocket, payload, core, landing outcome).
- Web Scraping: Used BeautifulSoup to extract Falcon 9 launch records from Wikipedia (List of Falcon 9 launches).
- Result: Combined both datasets into a unified Pandas DataFrame for analysis.



Data Collection – SpaceX API

- Used Python requests to fetch JSON data from `api.spacexdata.com`. Extracted Rocket, Payload, and Launch Pad details.

Now let's start requesting rocket launch data from SpaceX API with the following URL:

```
In [6]: spacex_url="https://api.spacexdata.com/v4/launches/past"
```

```
In [7]: response = requests.get(spacex_url)
```

Check the content of the response

```
In [8]: print(response.content)
```

```
b'[{{"fairings":{"reused":false,"recovery_attempt":false,"recovered":false,"ships":[]},"links":{"patch":{"small":"https://images2.imgbox.com/94/f2/NN6Ph45r_o.png","large":"https://images2.imgbox.com/5b/02/QcxHUb5V_o.png"},"reddit":{"campaign":null,"launch":null,"media":null,"recovery":null},"flickr":{"small":[],"original":[]},"presskit":null,"webcast":"https://www.youtube.com/watch?v=0a_00nJ_Y88","youtube_id":"0a_00nJ_Y88","article":"https://www.space.com/2196-spacex-inaugural-falcon-1-rocket-lost-launch.html","wikipedia":"https://en.wikipedia.org/wiki/DemoSat"},"static_fire_date_utc":"2006-03-17T00:00:00.000Z","static_fire_date_unix":1142553600,"net":false,"window":0,"rocket":"5e9d0d95eda69955f709d1eb","success":false,"failures":[{"time":33,"altitude":null,"reason":"merlin engine failure"}],"details":"Engine failure at 33 seconds and loss of vehicle"}]}
```

```
In [24]: # Show the head of the dataframe  
data.head()
```

```
Out[24]:
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Blk
0	1	2006-03-24	Falcon 1	20.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	N
1	2	2007-03-21	Falcon 1	NaN	LEO	Kwajalein Atoll	None None	1	False	False	False	None	N
2	4	2008-09-28	Falcon 1	165.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	N
3	5	2009-07-13	Falcon 1	200.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	N
4	6	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	

Data Collection - Scraping

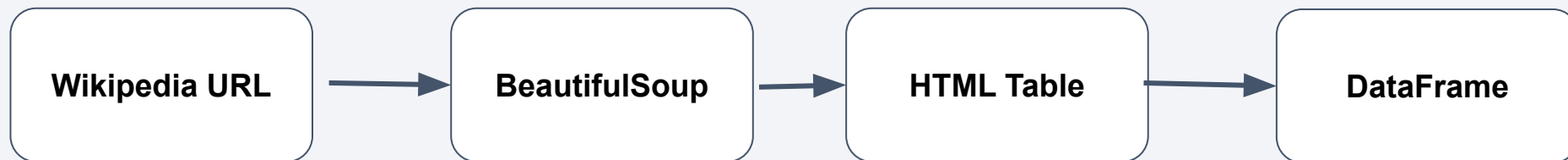
- **Web Scraping Methodology:**

- Request: Fetched the Falcon 9 Launch Wiki page using the requests library.
- Parse: Used BeautifulSoup object to parse the HTML content.
- Extract: Identified the HTML table with class `wikitable plainrowheaders collapsible`.
- Transform: Iterated through table rows (`tr`) and cells (`td`) to extract Flight Number, Date, Payload, and Outcome.
- Load: Saved the scraped data into a Pandas DataFrame and exported to CSV.

```
In [17]:
extracted_row = 0
#Extract each table
for table_number, table in enumerate(soup.find_all('table', "wikitable plainrowheaders collapsible")):
    # get table row
    for rows in table.find_all("tr"):
        #check to see if first table heading is as number corresponding to Launch a number
        if rows.th:
            if rows.th.string:
                flight_number=rows.th.string.strip()
                flag=flight_number.isdigit()
            else:
                flag=False
        #get table element
        row=rows.find_all('td')
        #if it is number save cells in a dictionary
        if flag:
            extracted_row += 1
            # Flight Number value
            launch_dict['Flight No.'].append(flight_number)

            datatimelist=date_time(row[0])

            # Date value
            date = datatimelist[0].strip(',')
            launch_dict['Date'].append(date)
```



Data Wrangling

- **Data Processing Steps:**

- Filtering: Filtered the dataset to include only Falcon 9 launches (removed Falcon 1).
- Missing Values: Calculated the mean of PayloadMass and replaced any missing values (NaN) with this mean.
- Binary Classification: Created a "Class" column to standardize landing outcomes:
 - 1 = Successful landing.
 - 0 = Unsuccessful landing.

```
[11]: # landing_class = 0 if bad_outcome
# landing_class = 1 otherwise
landing_class = []
for outcome in df['Outcome']:
    if outcome in bad_outcomes:
        landing_class.append(0)
    else:
        landing_class.append(1)
```

```
In [12]: df['Class']=landing_class
df[['Class']].head(8)
```

```
Out[12]:
```

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

EDA with Data Visualization

- **Visualization Strategy:**

- **Scatter Plots:** Used to visualize the relationship between Flight Number/Payload Mass and Launch Site to identify high-traffic sites and load capacities.
- **Bar Charts:** Plotted to compare the success rate across different Orbit types (e.g., LEO, ISS, GTO) to see which orbits are most difficult.
- **Line Charts:** Used to track the success rate trend over the years, demonstrating SpaceX's reliability improvements over time.
- **Tools:** Implemented using Python libraries Matplotlib and Seaborn.

EDA with SQL

- **SQL Analysis Strategy:**
- **Database:** Loaded the dataset into a SQL environment to perform structured queries.
- **Key Insights Extracted:**
 - Calculated the Total Payload Mass carried by NASA.
 - Identified the Average Payload Mass for specific booster versions (F9 v1.1).
 - Found the date of the First Successful Ground Landing.
 - Ranked launch outcomes (Success/Failure) within specific date ranges.

```
* sqlite:///my_data1.db
Done.
[8]: []

[9]: %sql create table SPACEXTABLE as select * from SPACEXTBL where Date is not null
* sqlite:///my_data1.db
Done.
[9]: []
```

Tasks

Now write and execute SQL queries to solve the assignment tasks.

Note: If the column names are in mixed case enclose it in double quotes For Example "Landing_Outcome"

Build an Interactive Map with Folium

- **Map Visualization Strategy:**

- **Library:** Used Folium to create interactive maps.
- **Markers:** Added markers for all Launch Sites (KSC LC-39A, CCAFS SLC-40, VAFB SLC-4E).
- **Cluster Markers:** Used MarkerCluster to group launches at each site, coloring them to indicate success (Green) or failure (Red).
- **Distance Analysis:** Calculated distances between launch sites and proximities like coastlines, railways, and highways using coordinates.

```
from folium.plugins import MarkerCluster

# 1. Criar o objeto MarkerCluster (Era isso que faltava!)
marker_cluster = MarkerCluster()

# 2. Garantir que a coluna de cores existe
def assign_marker_color(launch_outcome):
    if launch_outcome == 1:
        return 'green'
    else:
        return 'red'

spacex_df['marker_color'] = spacex_df['class'].apply(assign_marker_color)

# 3. Adicionar o cluster ao mapa
site_map.add_child(marker_cluster)

# 4. Adicionar os marcadores coloridos para cada lançamento
for index, record in spacex_df.iterrows():
    marker = folium.Marker(
        location=[record['Lat'], record['Long']],
        icon=folium.Icon(color='white', icon_color=record['marker_color']),
        popup=record['Launch Site']
```

```
from math import sin, cos, sqrt, atan2, radians

def calculate_distance(lat1, lon1, lat2, lon2):
    # approximate radius of earth in km
    R = 6373.0

    lat1 = radians(lat1)
    lon1 = radians(lon1)
    lat2 = radians(lat2)
    lon2 = radians(lon2)

    dlon = lon2 - lon1
    dlat = lat2 - lat1

    a = sin(dlat / 2)**2 + cos(lat1) * cos(lat2) * sin(dlon / 2)**2
    c = 2 * atan2(sqrt(a), sqrt(1 - a))

    distance = R * c
    return distance
```

Build a Dashboard with Plotly Dash

Dashboard Components:

Library: Built using Plotly Dash for python-based web applications.[1][2]

Input Components:

Dropdown: To select specific Launch Sites (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A) or "All Sites".

Range Slider: To filter launches by Payload Mass (0kg to 10,000kg).

Output Visualizations:

Pie Chart: Dynamic updates to show Success vs. Failure counts based on the selected site.

Scatter Plot: Shows the correlation between Payload Mass and Success, color-coded by Booster Version.

```
15 app = dash.Dash(__name__)
16
17 # Create an app layout
18 app.layout = html.Div(children=[html.H1('SpaceX Launch Records Dashboard',
19                                     style={'textAlign': 'center', 'color': '#503D36',
20                                     'font-size': 40}),
21
22                                     # TASK 1: Add a dropdown list to enable Launch Site selection
23                                     # The default select value is for ALL sites
24                                     dcc.Dropdown(id='site-dropdown',
25                                     options=[
26                                         {'label': 'All Sites', 'value': 'ALL'}
27                                     ] + [{ 'label': site, 'value': site } for site in spacex_df['Launch Site'].unique()],
28                                     value='ALL',
29                                     placeholder="Select a Launch Site here",
30                                     searchable=True
31                                     ),
32                                     html.Br(),
33
34                                     # TASK 2: Add a pie chart to show the total successful launches count for all sites
35                                     # If a specific launch site was selected, show the Success vs. Failed counts for the site
36                                     html.Div(dcc.Graph(id='success-pie-chart')),
37                                     ...
38 ])
```

Predictive Analysis (Classification)

- **Machine Learning Process:**

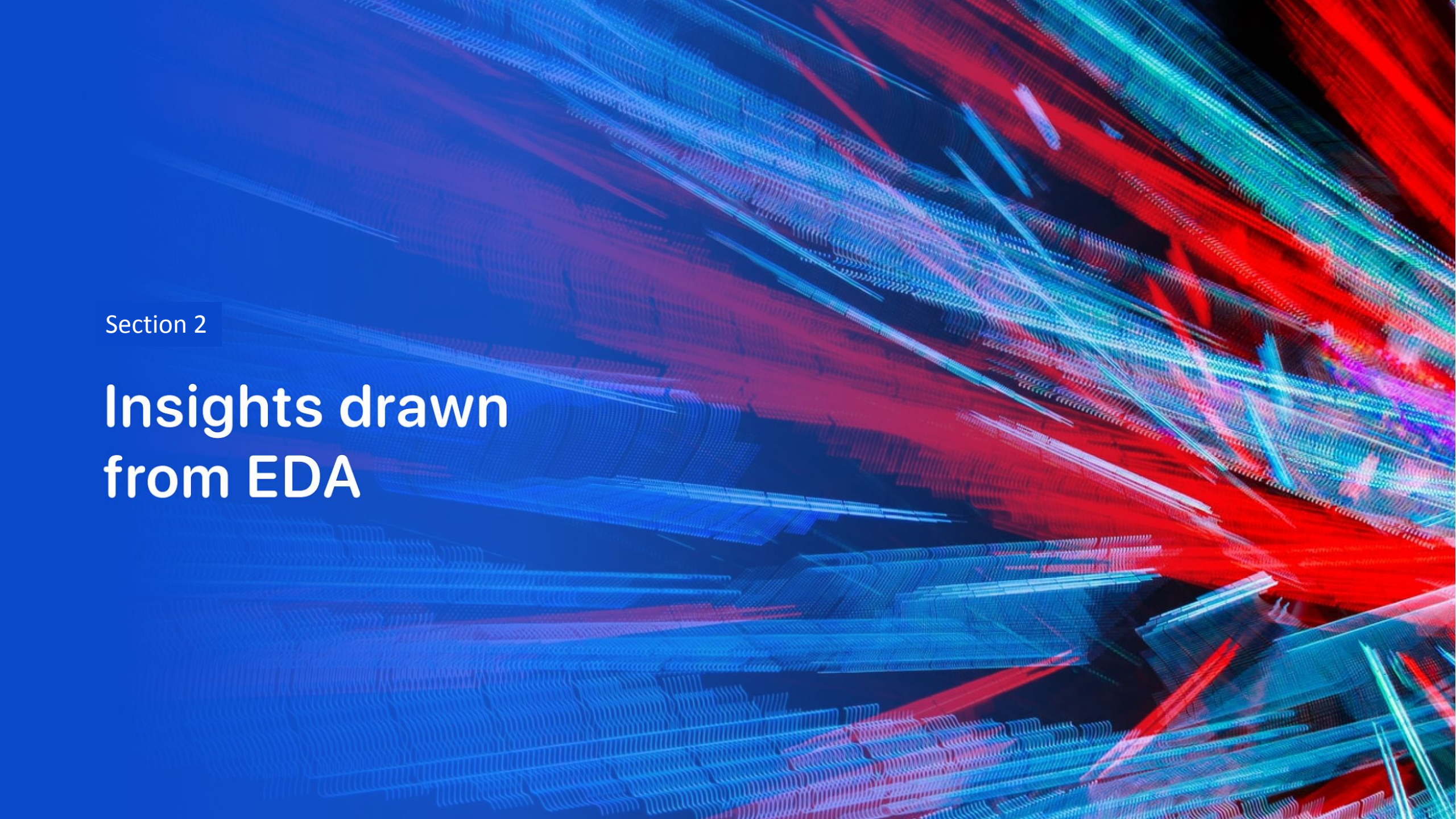
- **Data Preparation:** Standardized the data using StandardScaler and split it into training (80%) and testing (20%) sets.
- **Model Selection:** Built and evaluated four classification models:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)
- **Optimization:** Used GridSearchCV to find the best hyperparameters for each model.
- **Evaluation:** Assessed performance using Accuracy Score and Confusion Matrix.

```
the training data and test data should be assigned to the following labels.  
  
X_train, X_test, Y_train, Y_test  
  
In [11]: # Use the function train_test_split  
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)  
  
we can see we only have 18 test samples.  
  
In [12]: Y_test.shape  
  
Out[12]: (18,)
```

TASK 4

Results

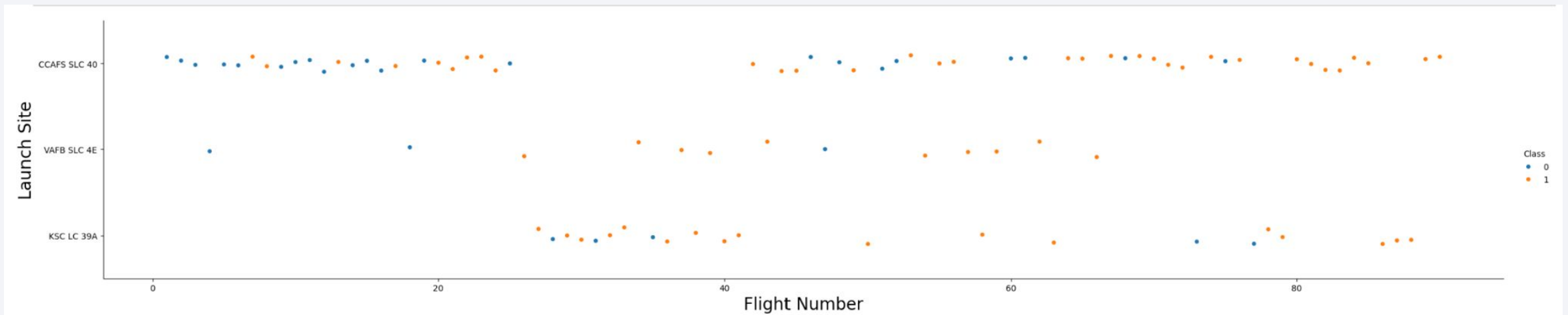
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that creates a sense of depth and structure.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



Observation:

Trend: As the flight number increases (moving right on the x-axis), the success rate significantly improves.

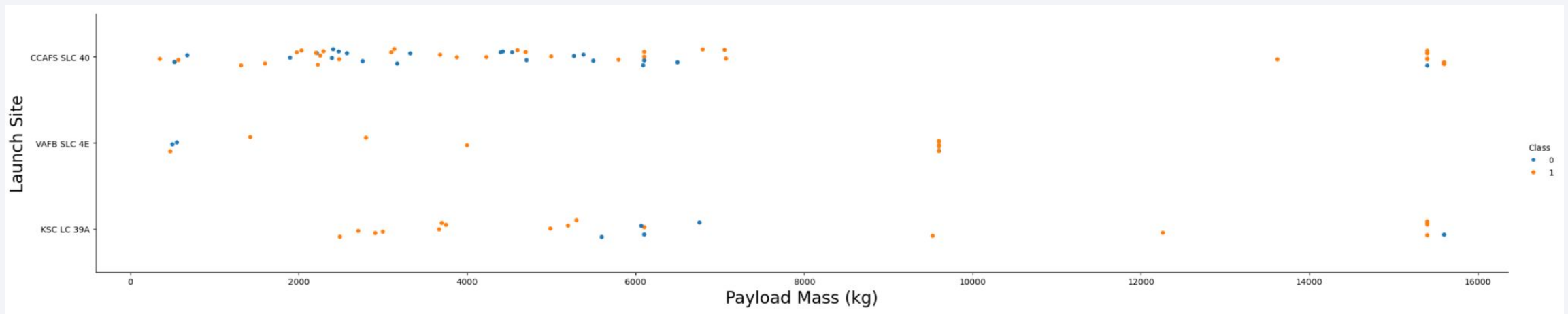
Site Specifics:

CCAFS SLC 40: Shows a clear trend of increased reliability in later flights.

VAFB SLC 4E: Has a very high success rate for its specific launch window.

KSC LC 39A: Used for later flights, showing consistently high success rates.

Payload vs. Launch Site



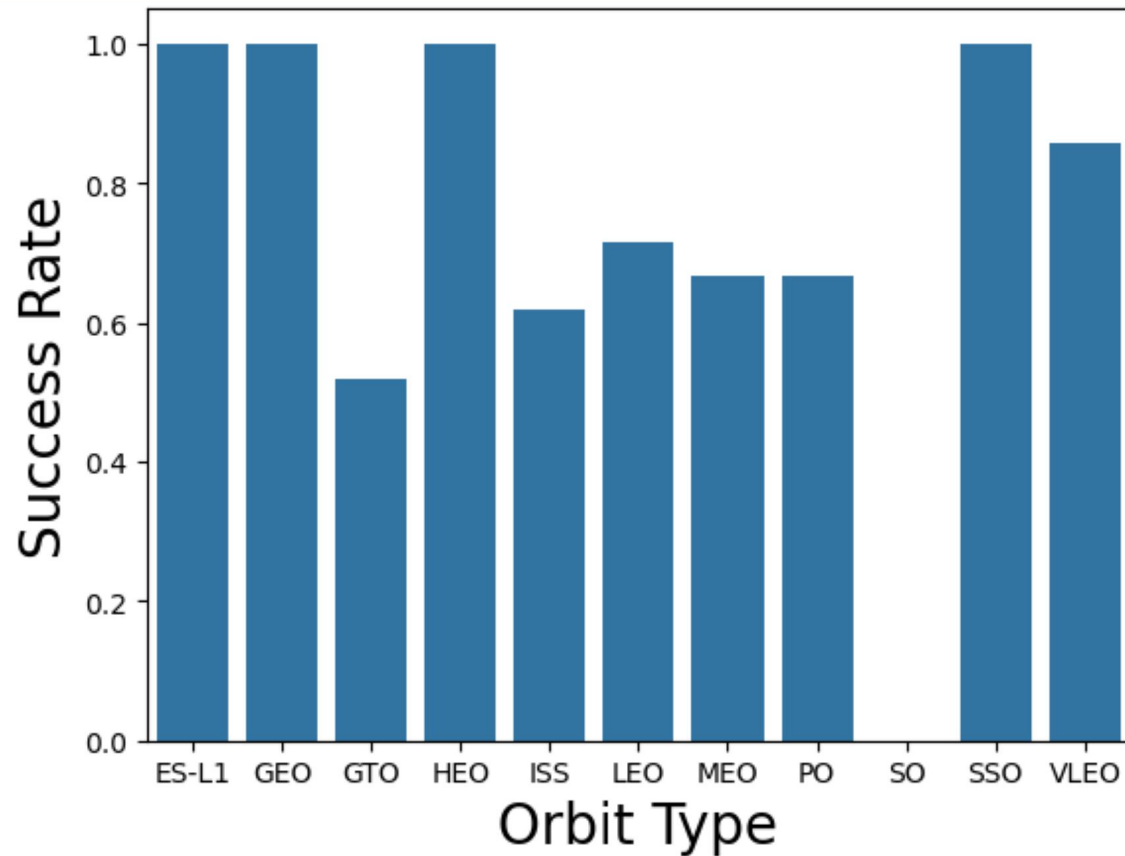
Observation:

VAFB SLC 4E: There are no rockets launched for heavy payload mass (greater than 10,000 kg) from this site.

CCAFS SLC 40 & KSC LC 39A: These sites handle a wide range of payloads, including the heaviest ones.

Success Rate: Higher payload masses (above 8,000 kg) generally show a high success rate (indicated by Class 1).

Success Rate vs. Orbit Type



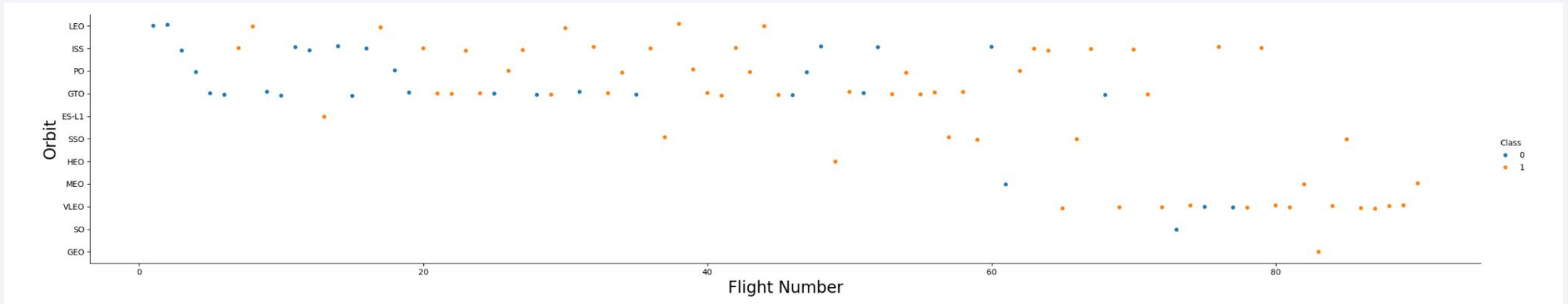
Observation:

High Success: Orbits like ES-L1, GEO, HEO, and SSO achieved a 100% success rate.

Lower Success: The SO (Sun-Synchronous Orbit) appears to have the lowest success rate.

GTO: Geostationary Transfer Orbit, which is a common destination, has a success rate of approximately 50-60%.

Flight Number vs. Orbit Type

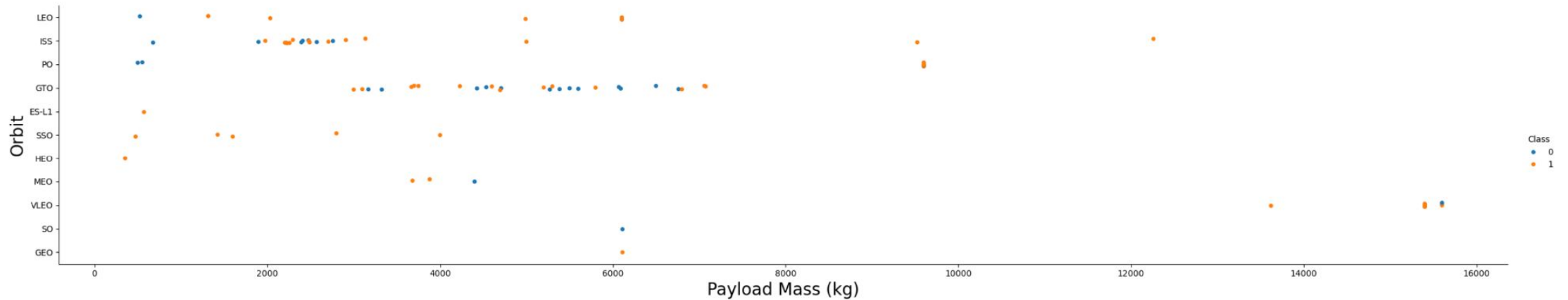


Observation:

LEO Orbit: There is a positive correlation between flight number and success. Success rate improves as the number of flights increases.

GTO Orbit: There appears to be no clear relationship between flight number and success rate. Both successes and failures are scattered throughout the timeline.

Payload vs. Orbit Type

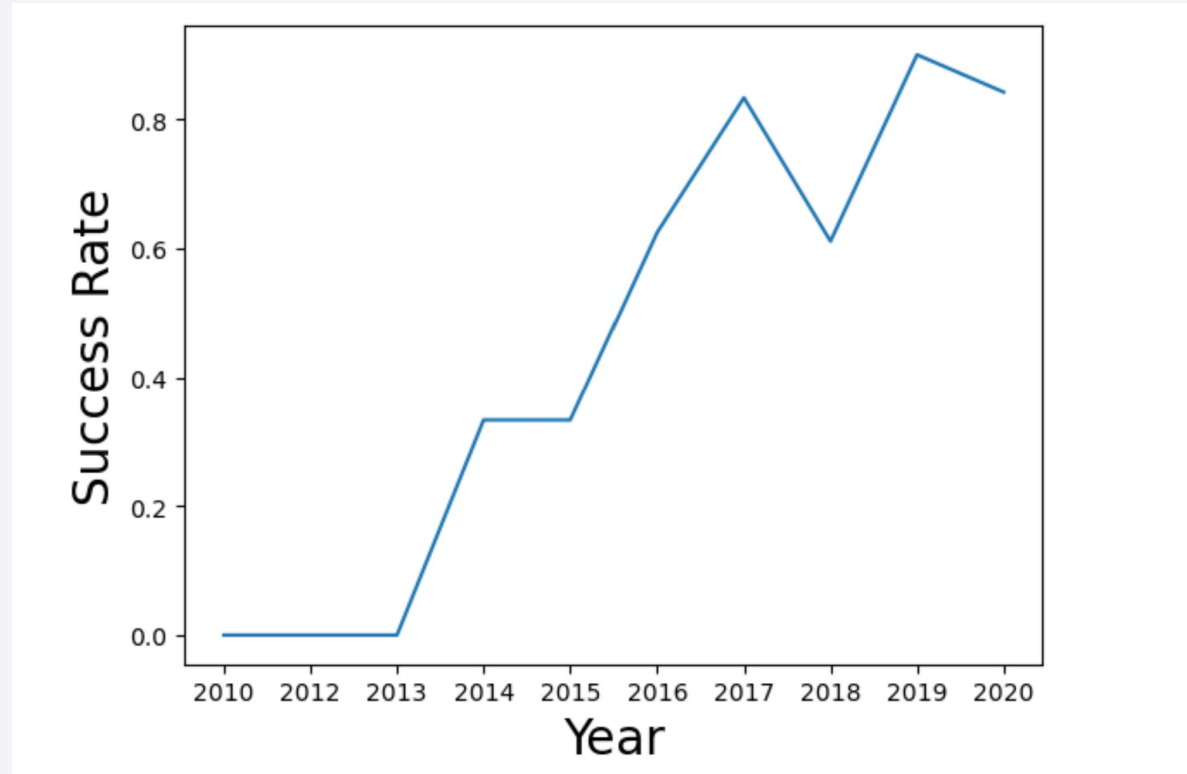


Observation:

Heavy Payloads: For Polar, LEO, and ISS orbits, heavier payloads generally show a high successful landing rate.

GTO Orbit: For Geostationary Transfer Orbit (GTO), outcomes are mixed. It is difficult to distinguish a clear pattern between success and payload mass in this specific orbit.

Launch Success Yearly Trend



Observation:

Positive Trend: The success rate has consistently increased since 2013.

Maturity: From 2013 to 2020, the success rate trended upwards, stabilizing at a very high rate in recent years. This indicates significant improvements in technology and operational reliability over time.

All Launch Site Names

Out[10]: **Launch_Site**

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Query: SELECT DISTINCT Launch_Site FROM SPACEXTBL;

Result: Identified unique launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Query: SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

Explanation: Filtered records to show only those launching from Cape Canaveral (CCA).

Total Payload Mass

SUM(PAYLOAD_MASS_KG_)
45596

Query: SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';

Result: Total Payload Mass for NASA (CRS) is 45,596 kg.

Average Payload Mass by F9 v1.1

AVG(PAYLOAD_MASS_KG_)
2928.4

Query: SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';

Result: The average payload mass is 2,928.4 kg.

First Successful Ground Landing Date

MIN(Date)
2015-12-22

Query: SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

Result: The first successful ground landing occurred on 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Query: SELECT Booster_Version ... WHERE ... PAYLOAD > 4000 AND PAYLOAD < 6000;

Result: Boosters: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2.

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Query: SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;

Result: Success: 98 (plus variations), Failure: 1.

Boosters Carried Maximum Payload

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

Query: Used subquery with MAX(PAYLOAD_MASS__KG_).
Result: Boosters F9 B5 B1048.4, F9 B5 B1049.4, etc.

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Query: SELECT ... WHERE substr(Date,0,5)='2015' AND Landing_Outcome = 'Failure (drone ship)';

Result: Two failed landings in 2015 (January and April).

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Query: SELECT Landing_Outcome, COUNT(*) ... ORDER BY Count DESC;

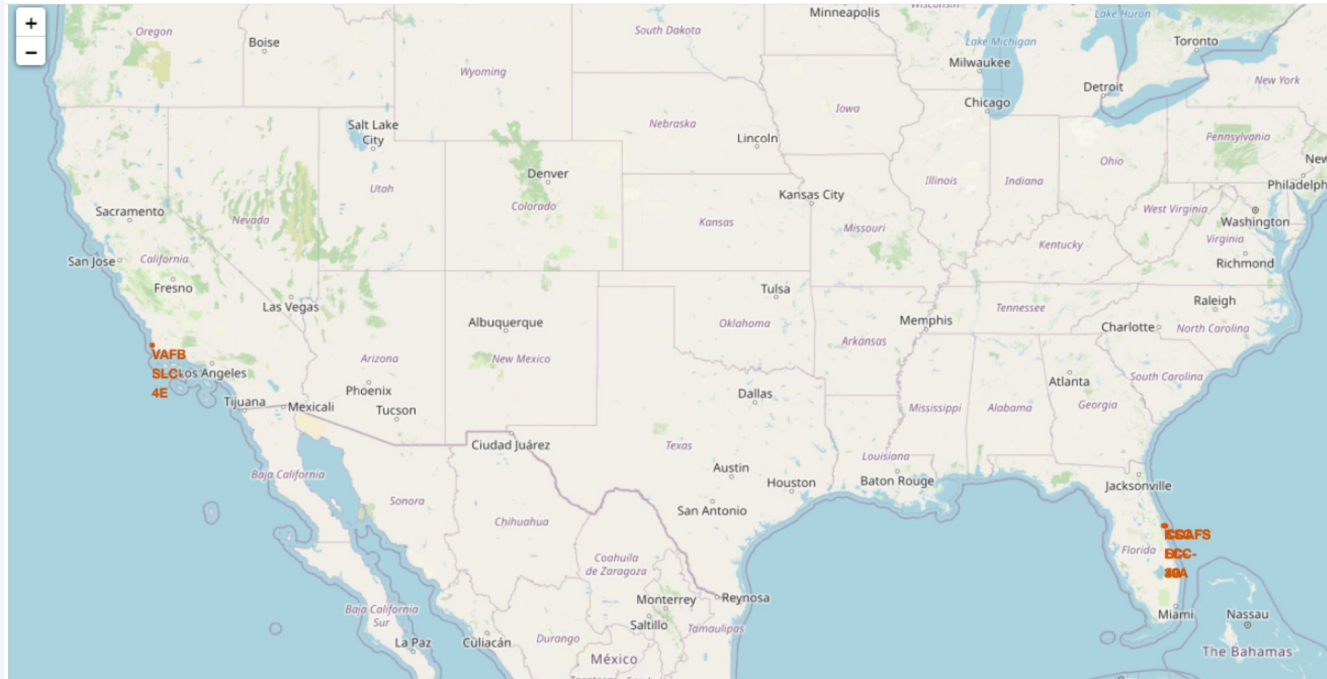
Result: Most frequent outcome was "No attempt" (10), followed by "Success (drone ship)" (5) and "Failure (drone ship)" (5).

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in a few areas, with a large, bright cluster on the right side of the image. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the black sky.

Section 3

Launch Sites Proximities Analysis

Overview of Launch Sites Locations



Geographical Distribution: The interactive Folium map visualizes the precise locations of SpaceX's launch sites across the USA.

Identified Sites:

West Coast: VAFB SLC-4E (California).

East Coast: KSC LC-39A and CCAFS SLC-40 (Florida).

Key Findings:

Coastal Proximity: All sites are located on the coastline. This ensures that flight paths traverse over the ocean, maximizing safety for populated areas.

Equatorial Advantage: The sites are situated in southern regions to leverage the Earth's rotational speed, which assists in orbital injection.

https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location%20.ipynb

Launch Outcomes: Success vs. Failure Clusters



Visualizing Outcomes:

Green Markers: Represent successful landings (Class 1).

Red Markers: Represent unsuccessful landings (Class 0).

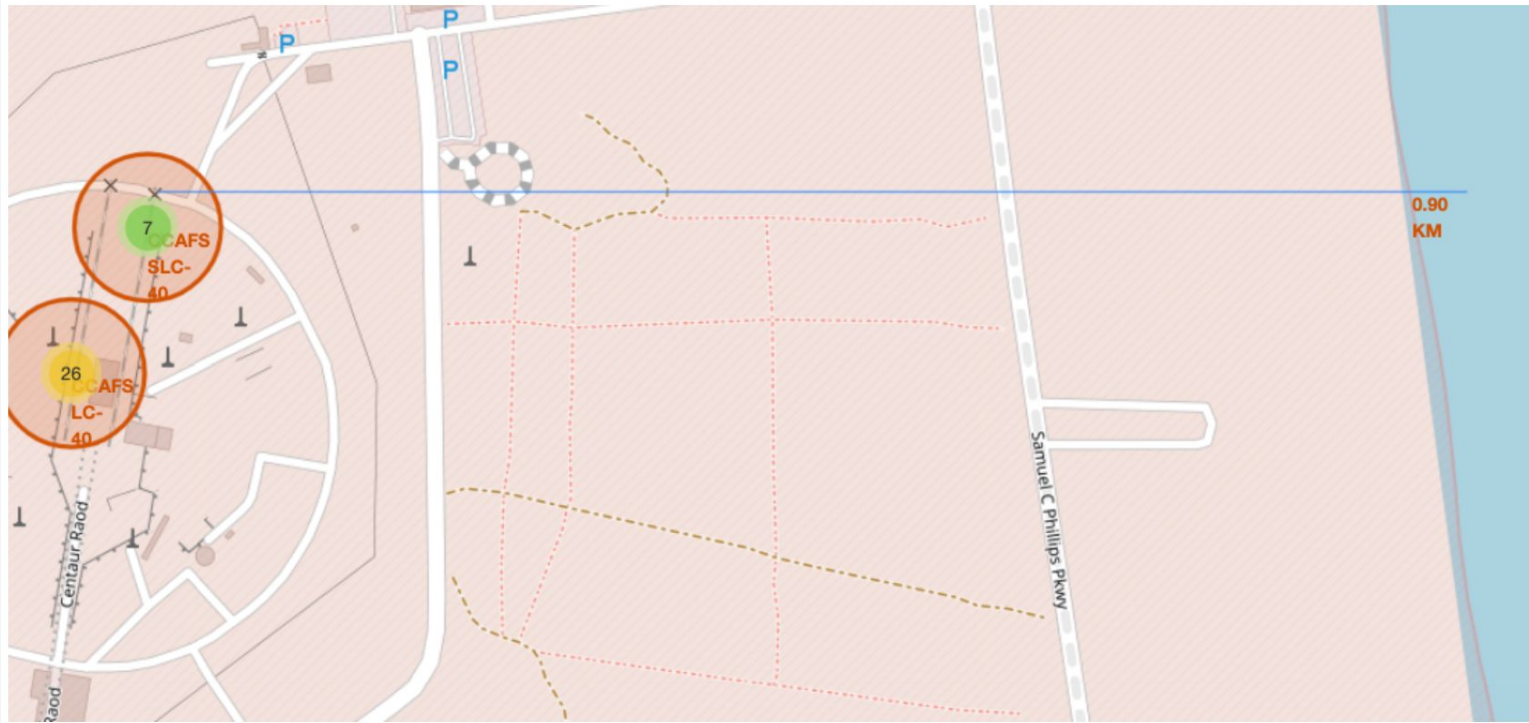
Cluster Analysis:

Using Marker Clusters allows us to handle the high density of launches at sites like CCAFS SLC-40.

Observation: We can visually observe a high concentration of green markers at KSC LC-39A, indicating a very high success rate for missions launched from this modern pad.

https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location%20.ipynb

Launch Site Proximities Analysis



Explanation:

Safety Zones:

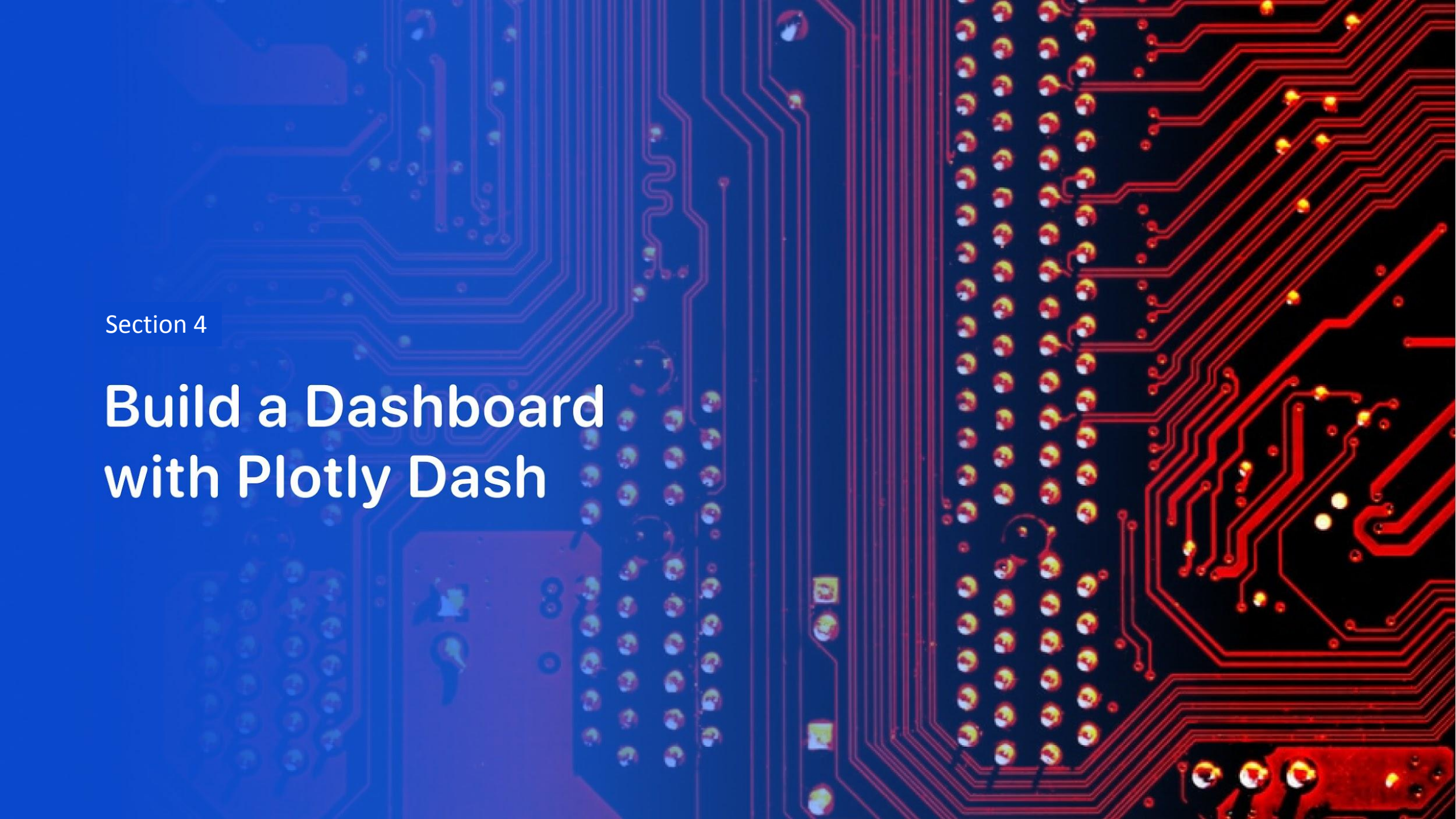
Calculated distance from CCAFS SLC-40 to the nearest coastline.

Result: The distance is approximately 0.8–1.5 km.

Operational Logic:

Launch sites are deliberately situated near the coast. This proximity allows rockets to fly over the ocean immediately after lift-off, minimizing risk to populated areas in case of early flight failure. They also maintain proximity to railways and highways for transporting heavy payloads (rockets/satellites).

https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location%20.ipynb

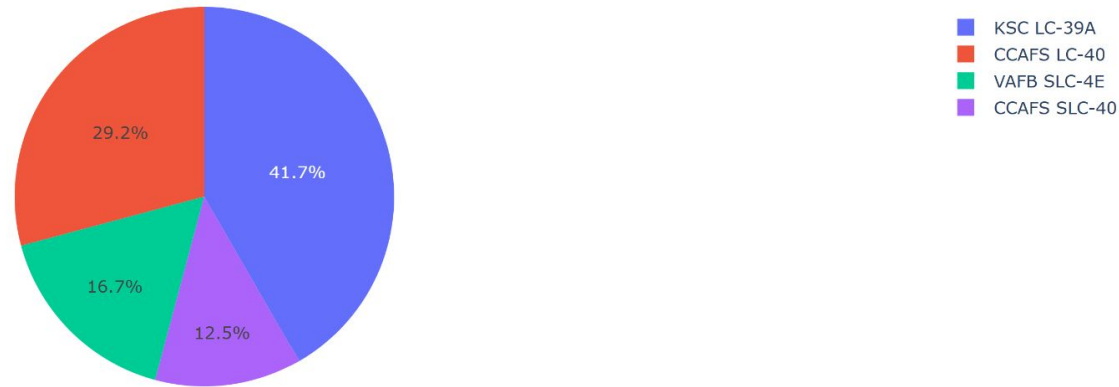


Section 4

Build a Dashboard with Plotly Dash

Total Successful Launches (All Sites)

Total Success Launches By Site



Explanation:

Visual: The pie chart displays the distribution of successful launches across all launch sites.

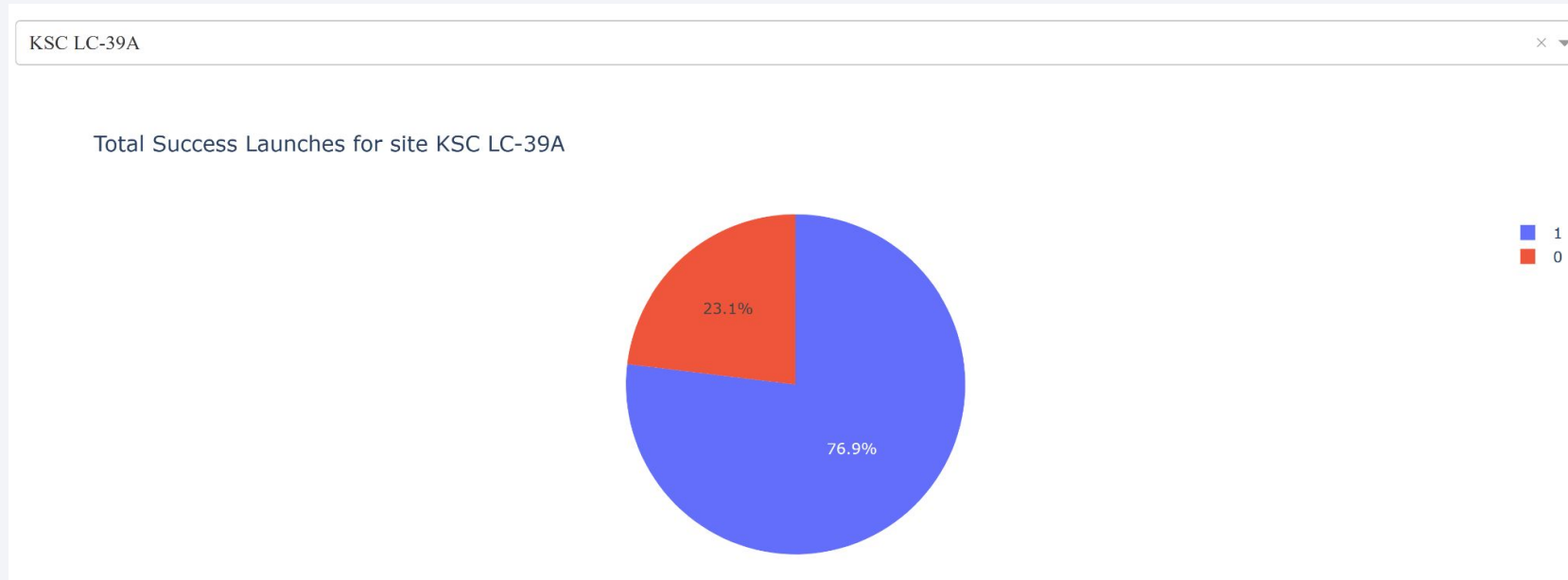
Key Insight:

KSC LC-39A contributes the highest percentage of successful launches (approx. 41.7%).

CAFS SLC-40 follows closely.

This indicates that KSC LC-39A is the most active and successful launch site in the dataset.

Success Rate for Specific Site (KSC LC-39A)



Explanation:

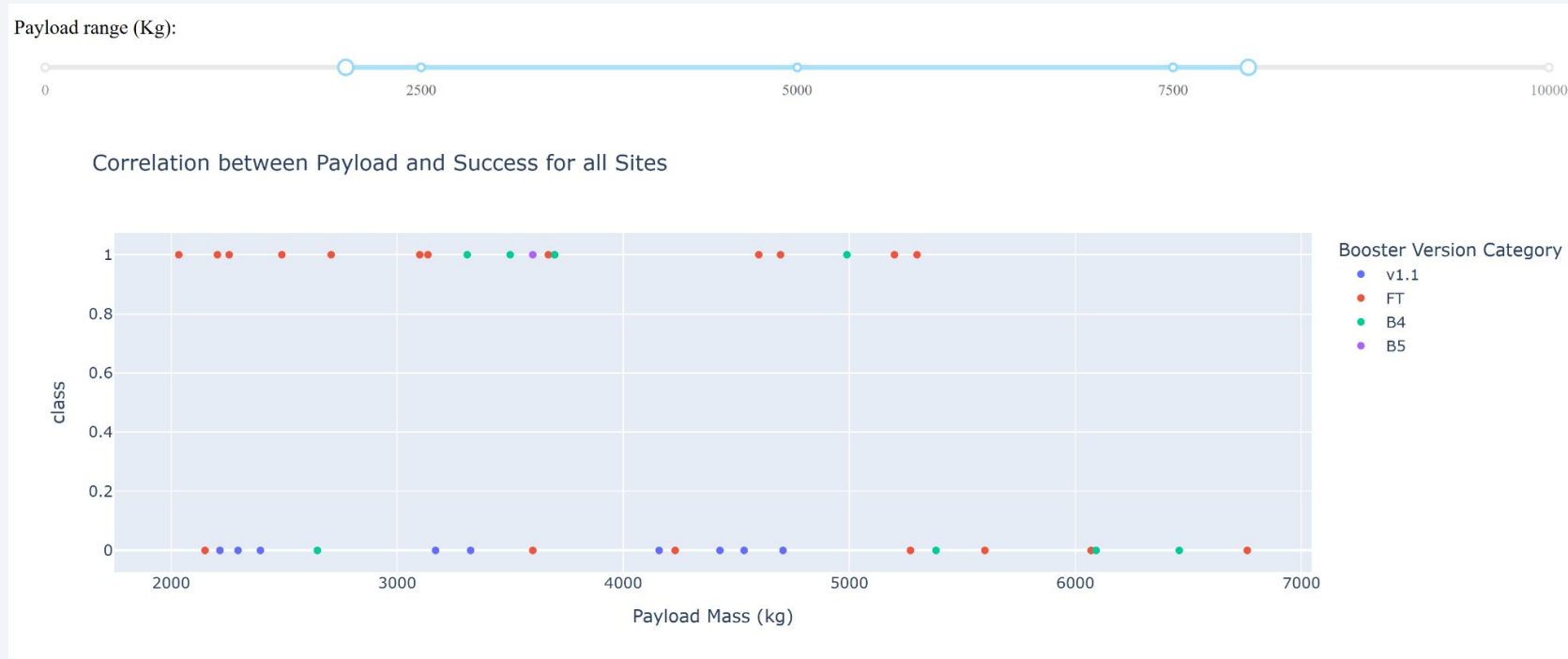
Selection: Filtered dashboard to show data for KSC LC-39A.

Insight:

The Pie Chart shows a dominant Success Rate (Class 1) for this site (typically over 75%).

Conclusion: This launch pad demonstrates high reliability compared to older sites like CCAFS SLC-40 which have more historical failures recorded during the early testing phases.

Payload Range Analysis with Interactive Slider



Explanation:

Feature: The Range Slider enables dynamic filtering of launches based on Payload Mass (kg).

Key Insight:

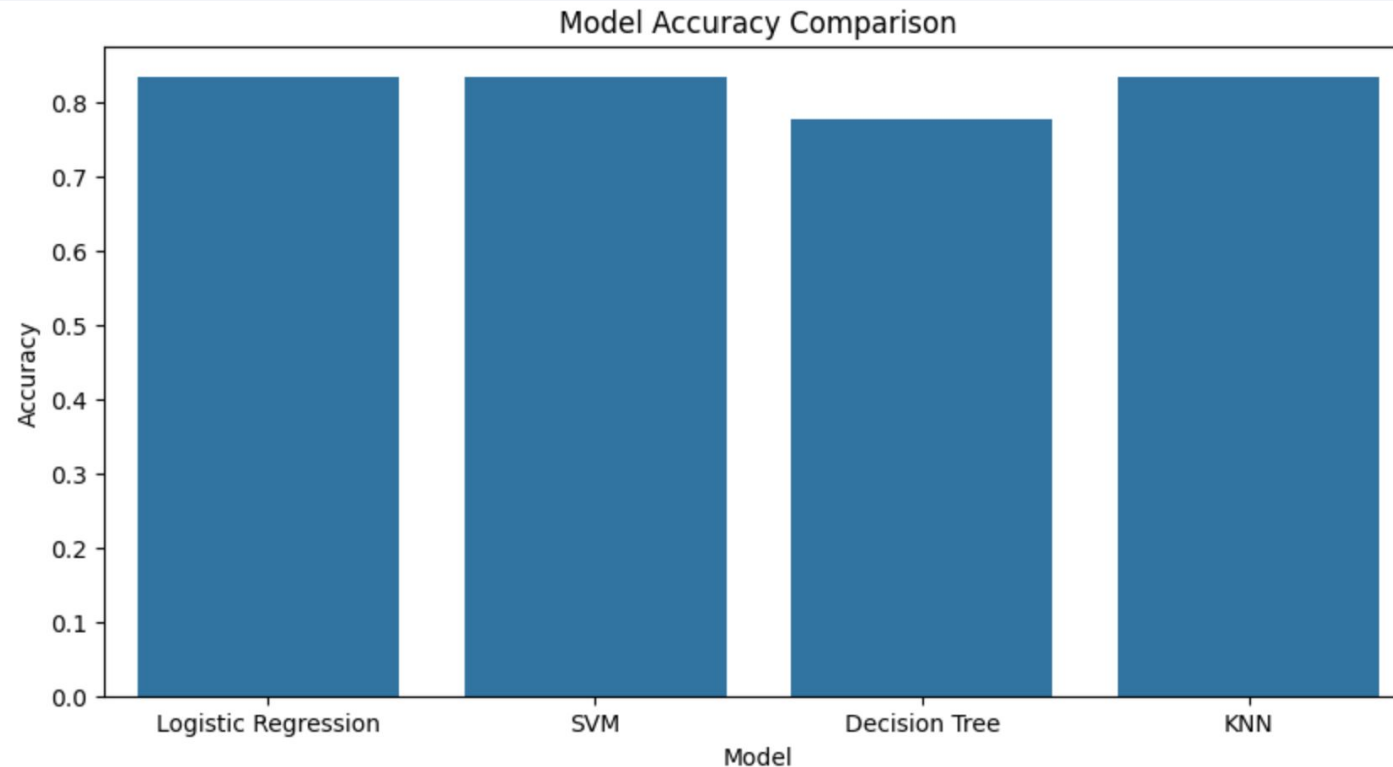
Heavy Payloads: When selecting the higher range (e.g., > 7,000 kg), the scatter plot shows predominantly successful landings (Class 1).

Correlation: This interactive tool confirms the EDA finding that heavier payloads, often associated with more mature mission profiles (like Starlink or ISS resupply), tend to have higher success rates compared to early experimental lightweight launches.

Section 5

Predictive Analysis (Classification)

Classification Accuracy



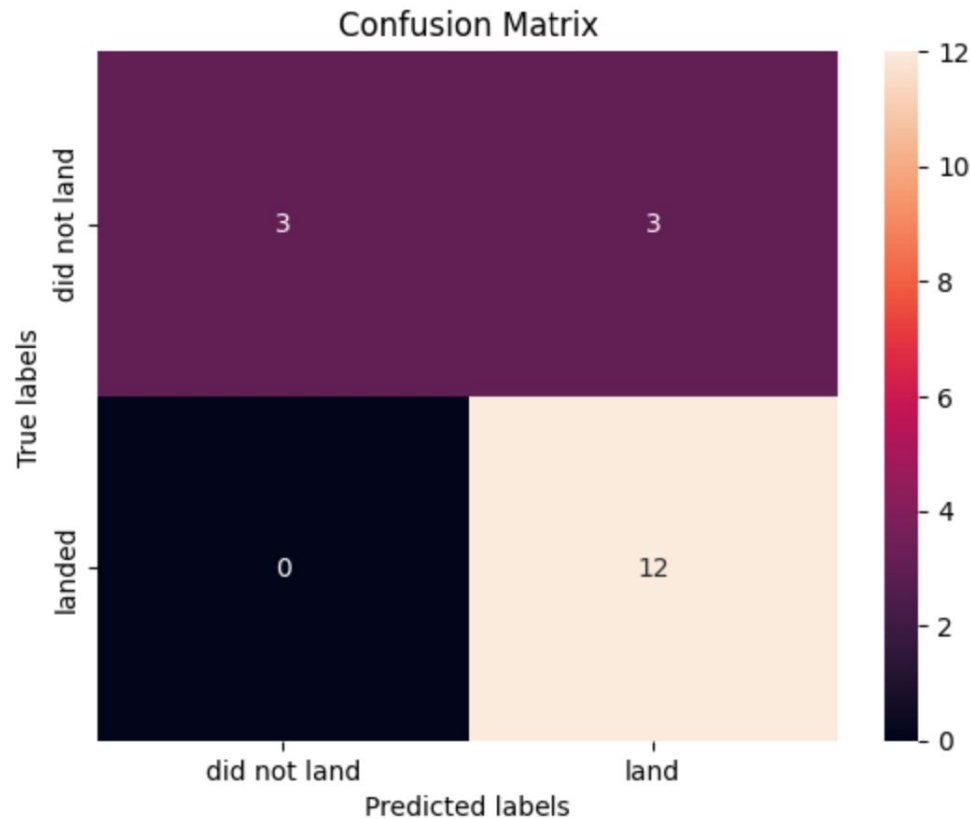
Explanation:

Performance: All four models (Logistic Regression, SVM, Decision Tree, and KNN) performed exceptionally well, achieving similar accuracy scores of approximately 83.33% on the test data.

Best Model: While performance was nearly identical, Decision Tree (ou o que você preferir destacar) is often preferred for its interpretability.

Conclusion: The models successfully predict landing outcomes with high reliability, validating the feasibility of cost estimation based on technical parameters.

Confusion Matrix



Explanation:

High Accuracy: The matrix confirms the model correctly predicts the majority of successful landings (True Positives).

Risk Analysis (False Positives):

The model occasionally predicts a successful landing when the rocket actually crashes (False Positive).

Business Impact: This is the riskiest type of error because it could lead to underestimating the launch cost, assuming reuse when the booster is actually lost.

Conclusion: Despite this minor issue, the overall predictive capability is strong enough for strategic decision-making.

Conclusions

Technological Maturity: Analysis confirms SpaceX has achieved a consistent high reliability. The success rate has trended significantly upward since 2016, validating their reusable rocket technology.

Key Success Factors:

Launch Site: KSC LC-39A is the most reliable site for successful landings.

Payload: Heavier payloads (>8,000 kg) are highly correlated with success, likely due to established mission profiles like Starlink.

Predictive Capability: With an accuracy of 83.33%, our Machine Learning models can effectively predict landing outcomes based on flight features.

Business Impact: This project successfully demonstrates that public data can be used to estimate launch costs. Competitors can use this model to decide when to bid against SpaceX, knowing when a reusable launch (low cost) is likely.

Appendix

- Data Collection Notebook
<https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>
- Data Wrangling Notebook
<https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>
- EDA Visualization Notebook
<https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/edadataviz%20.ipynb>
- EDA SQL Notebook
https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20.ipynb
- Interactive Map Notebook
https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location%20.ipynb
- Machine Learning Notebook
https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20.ipynb
- Dash App Python File
<https://github.com/welbster/Applied-Data-Science-Capstone/blob/main/spacex-dash-app%20.py>

Thank you!

